

Gate E4 report — run_v1.22_r1 (Paper IV v1.22-r1)

Verdict: **PASS — reused_verified_external_build**. Evidence: 20_EVIDENCE/E4/. Auditor: Claude Opus 5.5, same session as run_v1.2_r1/run_v1.21_r1; no Lean executed.

E4 — Formal build: documentary confirmation (no recompilation)

Status: **PASS — reused_verified_external_build**. No Lean, Lake, leanchecker or other build was run in this cycle (mandate §2). No pin, cache or compiled object was touched.

Why reuse is permitted (identity)

- The source cut is identical to the one built in run_v1.2_r1: 615/615 SOURCE_MANIFEST entries (fb459343...) match, with no extra files (E0 initial and final).
- The Lean source ZIP (cb274145...) and the annex ZIP (2847a422...) are unchanged (E0).
- The run_v1.2_r1 evidence is intact: its 1097 manifest files are unchanged on disk and the ZIP sidecar matches (E0).
- v1.22 changes the manuscript only, not Lean (E3).

Documentary re-verification

e4_reuse_check.py (the run_v1.21_r1 script, reused verbatim) → E4_REUSE_CHECK.json; the log is 10_LOGS/E4_reuse_check.log (see C2-06). By design it does **not** rely on E4_main/SUMMARY.json. Its sources are records.jsonl, the console log, the 608 module logs, the annex records, AuditorChecks.log, olean_vs_author.json, cache_diff.json and the source manifest.

Required evidence	Found
607 main modules	records.jsonl: 607/607 cut modules PASS (608 records including AuditorChecks); console modules_planned 607, pass 607, fail_or_blocked []; 607 PASS n 607 lines; EXIT 0; record source hashes equal the manifest for all 607
19 targets	console targets_status: all 19 FREEZE_SCOPE targets PASS
224 export controls	ReleaseExportCheck.lean has 224 #check; its log has 0 error lines
50 annex modules	annex records.jsonl: 50/50 PASS; BoundedCliqueGap axiom check: 633 declarations, axioms $\subseteq$ {propext, Classical.choice, Quot.sound}
Axioms and sorry	461 axiom records, none non-standard; 0 module logs with sorry text; the 1266-log detector scan has 0 hits (E8)
Byte identity and cache	607/607 oleans identical to the author's; auditor cone summary: 50 declarations, problems = 0; Mathlib cache unchanged (113821 files; no file changed, added or removed)

The resulting JSON is **identical** to the run_v1.21_r1 E4_REUSE_CHECK.json.

I reread the run_v1.2_r1 E4_RECORD.md. Its limitations remain in force: same machine and cache, a two-segment author build, and no kernel replay.